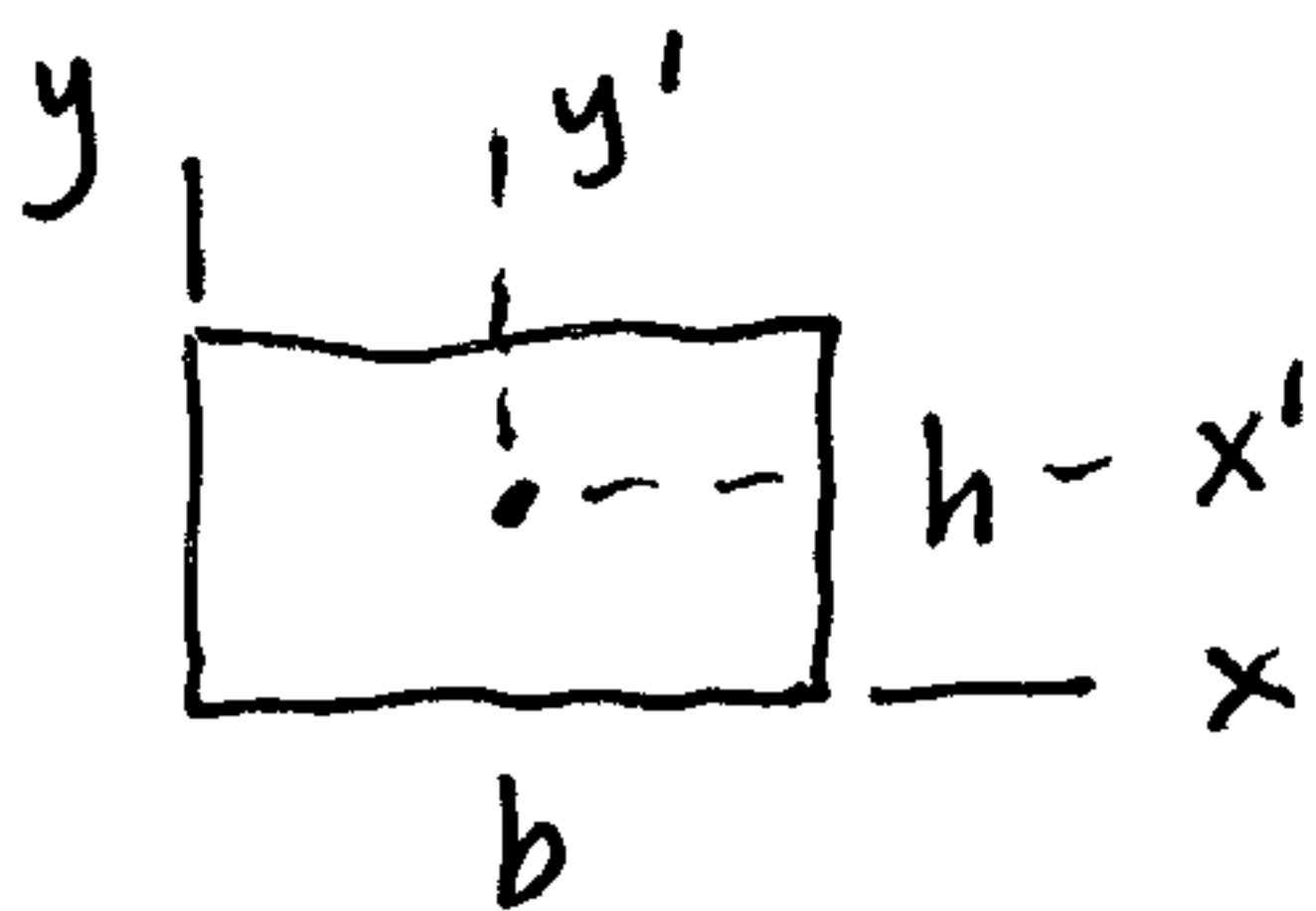
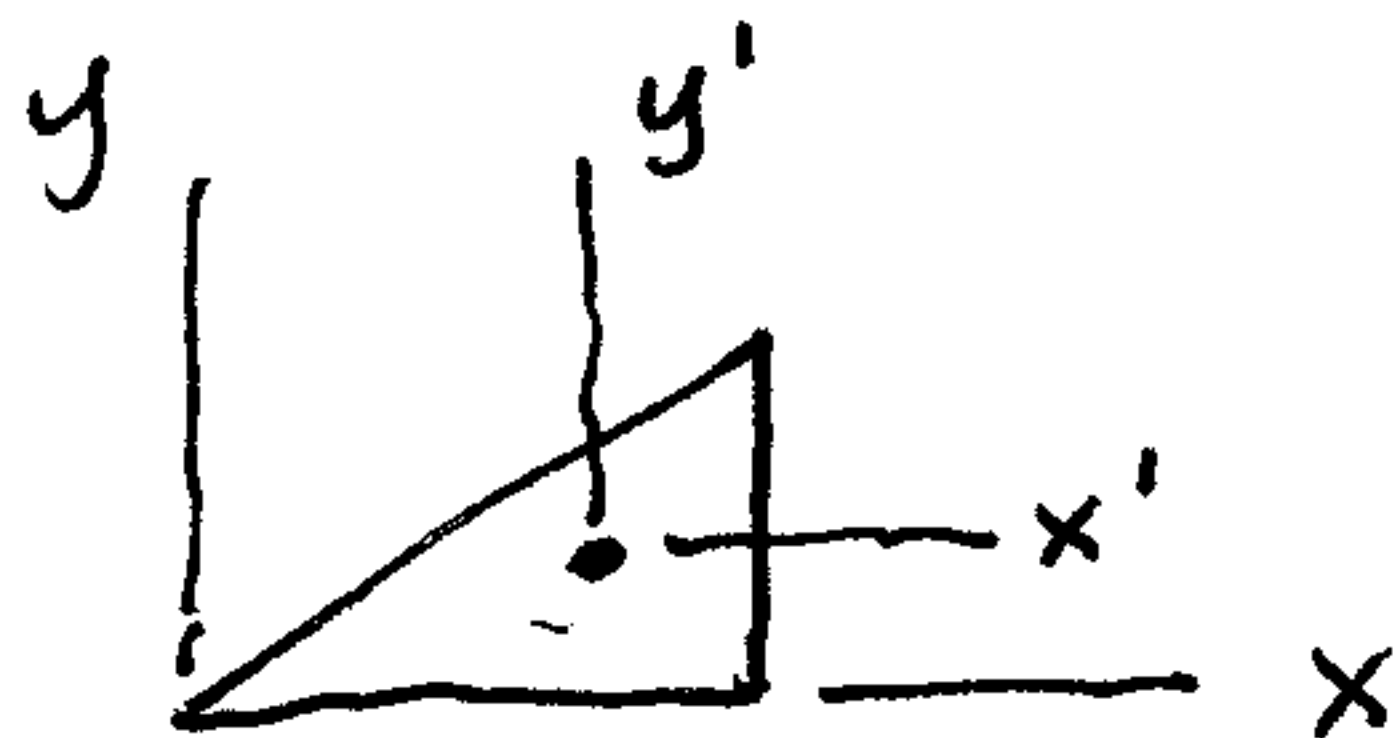


Hibbeler Crib Sheet



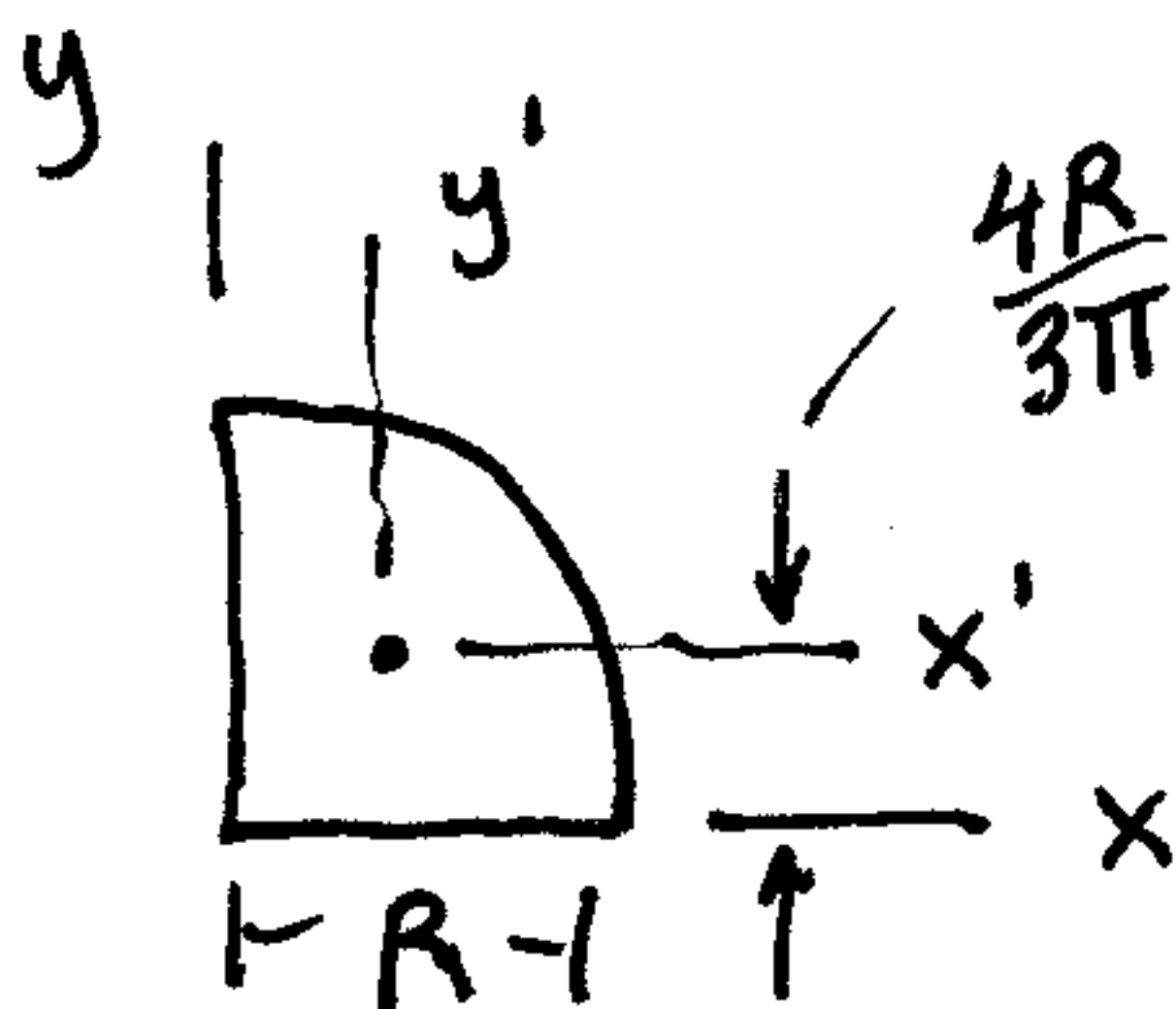
$$I_{xy} = \frac{1}{4} b^2 h^2$$

$$\bar{I}_{x'y'} = 0$$



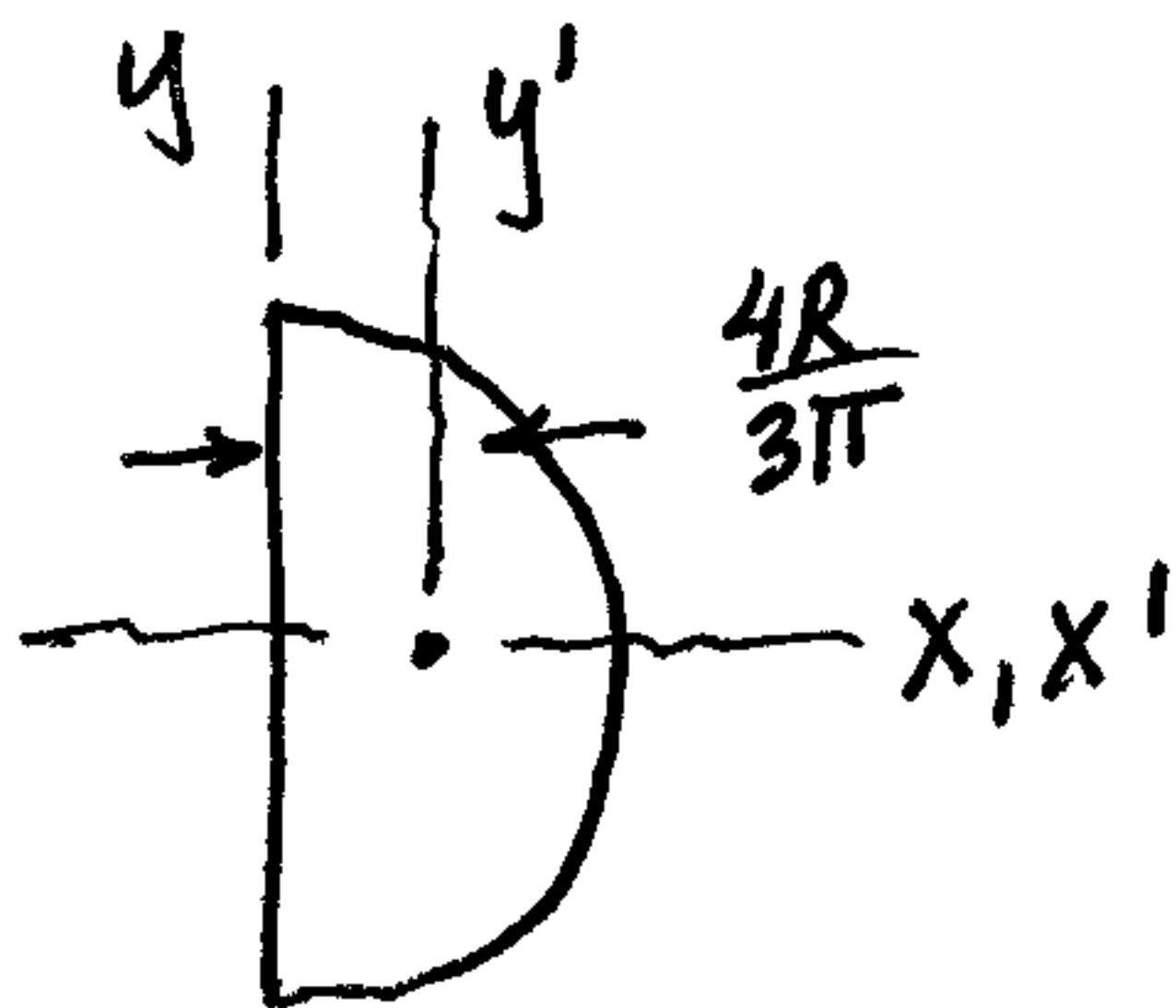
$$I_{xy} = \frac{1}{8} b^2 h^2$$

$$\bar{I}_{x'y'} = \frac{1}{72} b^2 h^2$$



$$I_{xy} = \frac{1}{8} R^4$$

No $\bar{I}_{x'y'}$ given.



$$I_{xy} = 0$$

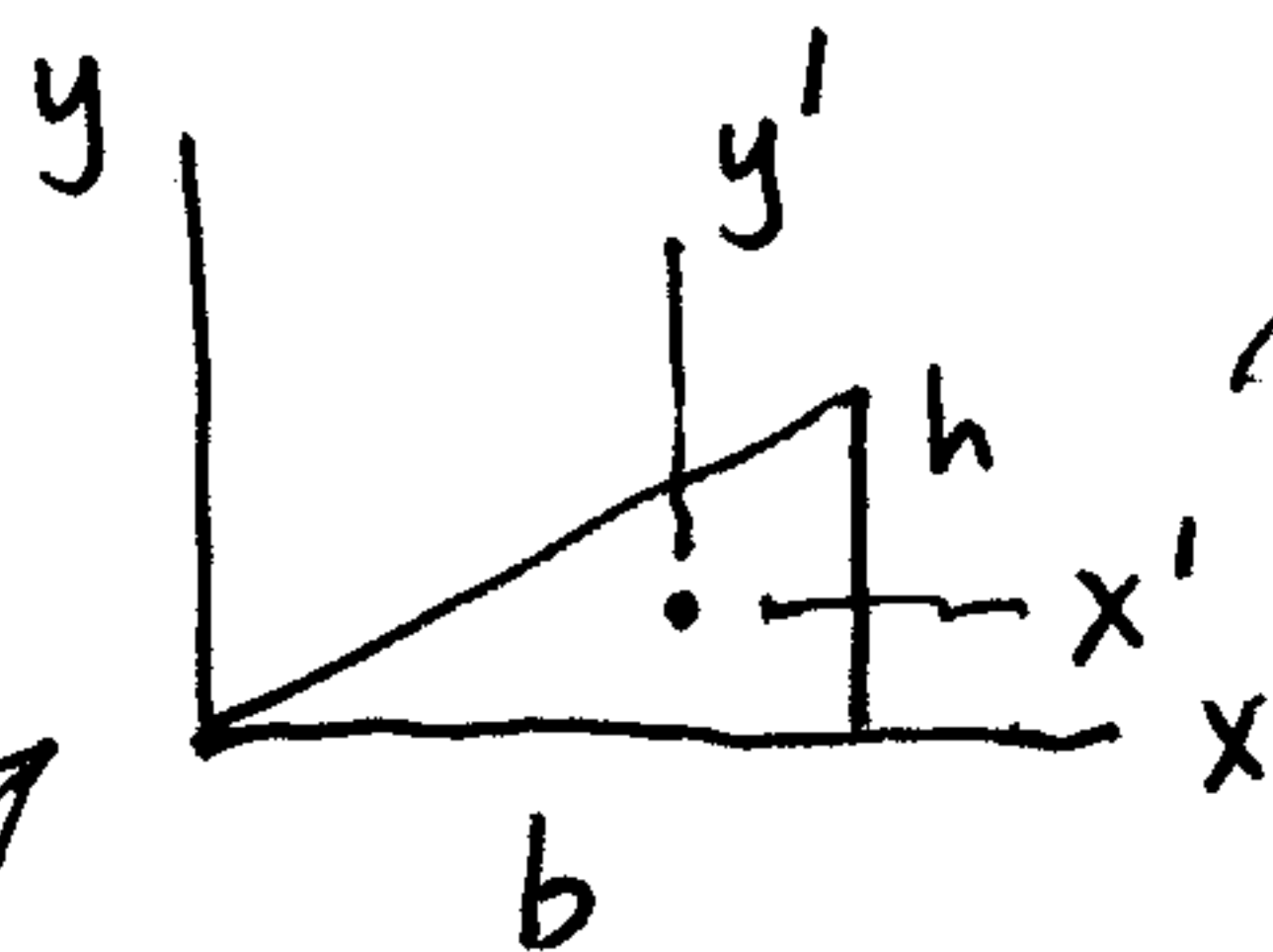
$$\bar{I}_{x'y'} = 0$$

① Rectangle: Given $\bar{I}_{x'y'} = 0$, find I_{xy}

$$\begin{aligned} \text{PAT } [I_{xy} &= \bar{I}_{x'y'} + A dx dy] \\ &= 0 + (b)(h) \left(\frac{b}{2}\right) \left(\frac{h}{2}\right) \end{aligned}$$

$$I_{xy} = \frac{1}{4} b^2 h^2$$

② Right Triangle

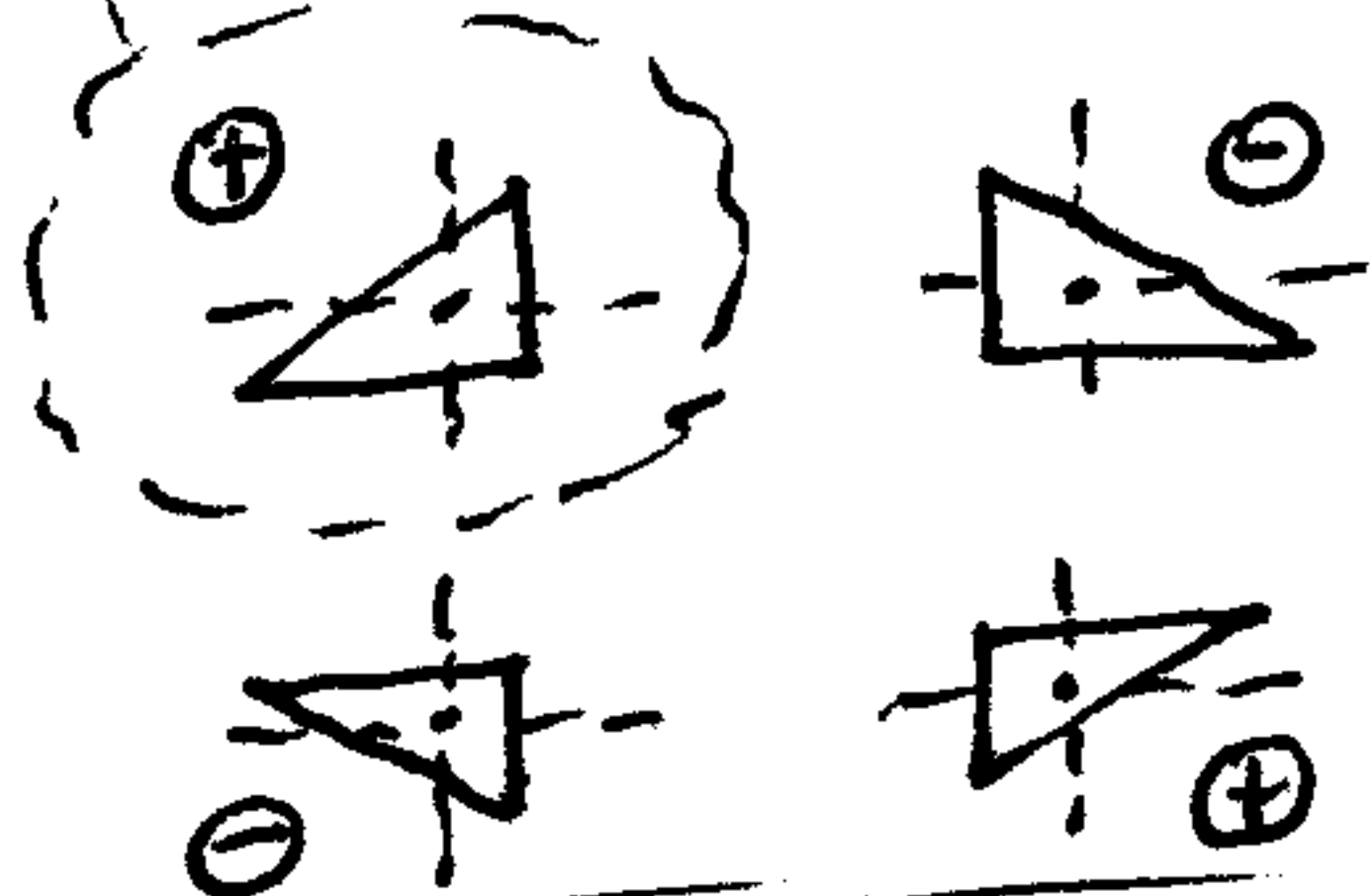


Given: $\bar{I}_{x'y'} = +\frac{1}{72} b^2 h^2$

Find I_{xy} :

$$\begin{aligned} I_{xy} &= \bar{I}_{x'y'} + A dx dy \\ &= \frac{1}{72} b^2 h^2 + \left(\frac{1}{2} b h\right) \left(\frac{2}{3} b\right) \left(\frac{1}{3} h\right) \\ &= \left(\frac{1}{72} + \frac{1}{9}\right) b^2 h^2 \\ &= \frac{8}{72} b^2 h^2 = \frac{1}{9} b^2 h^2 \end{aligned}$$

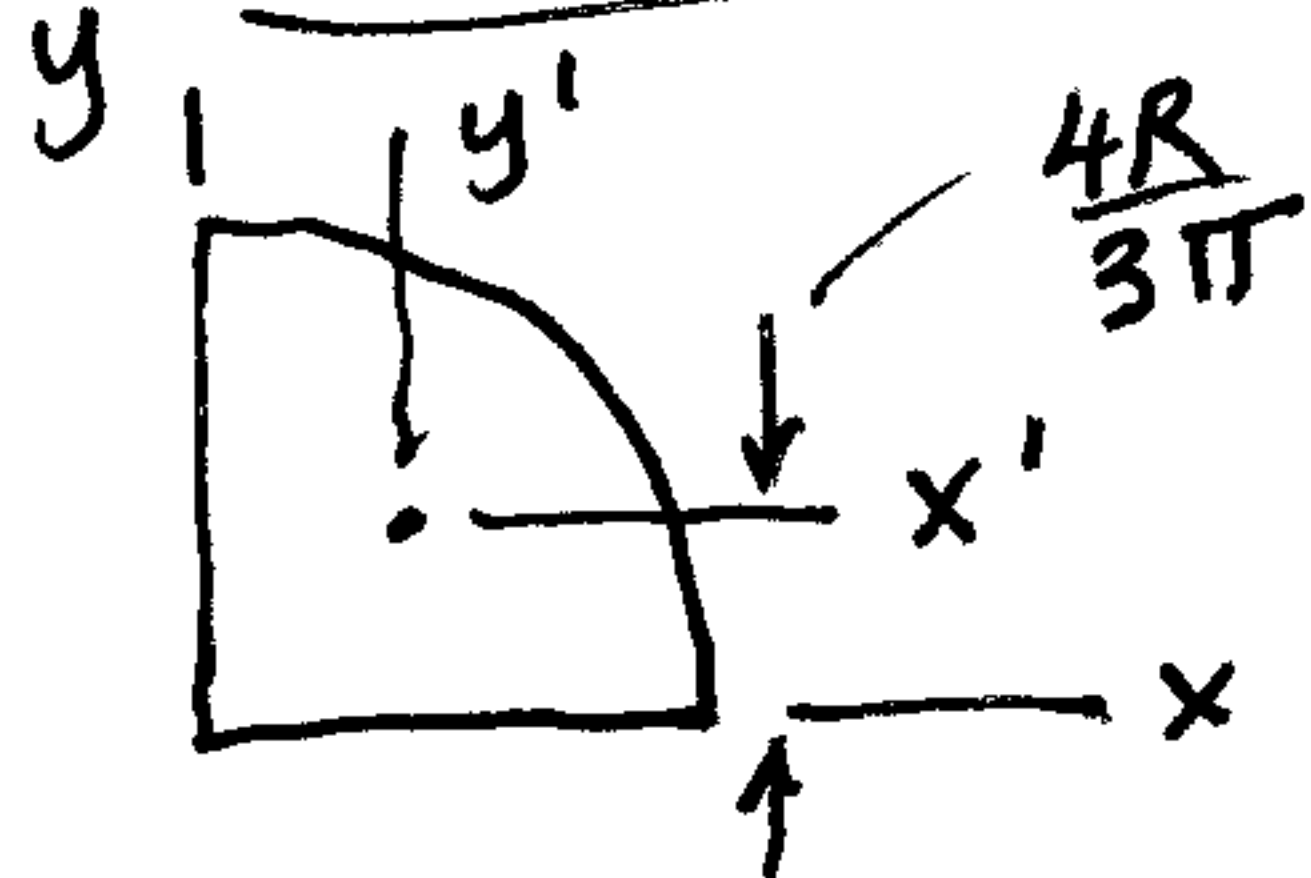
This $\bar{I}_{x'y'}$ value agrees w/ the Flori sheet:



$$\bar{I}_{x'y'} = (-1)^n \frac{1}{72} b^2 h^2$$

$$I_{xy} = \frac{1}{8} b^2 h^2$$

③ Quarter Circle



Hibbeler gives:

$$I_{xy} = \frac{1}{8} R^4$$

He does not give $\bar{I}_{x'y'}$.

Find $\bar{I}_{x'y'}$?

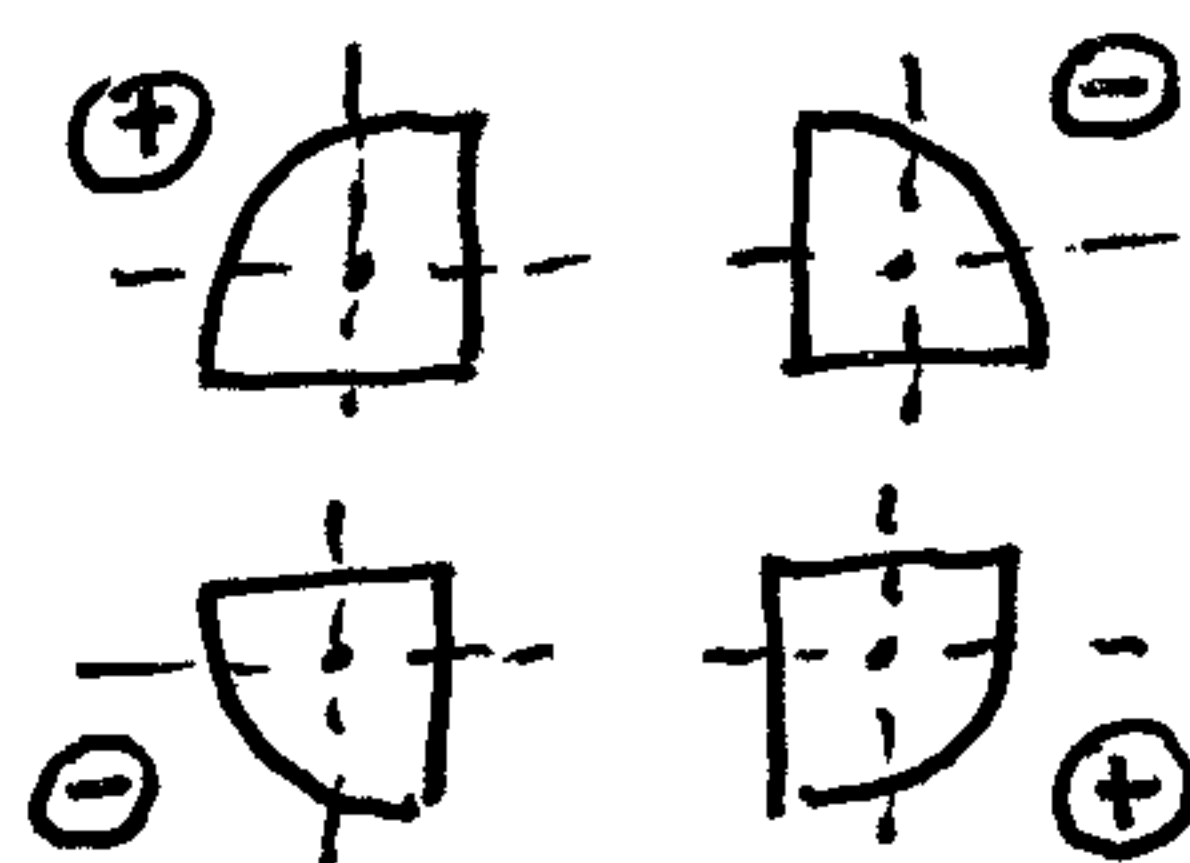
PAT: $I_{xy} = \bar{I}_{x'y'} + A dx dy$

$$\begin{aligned} \bar{I}_{x'y'} &= I_{xy} - A dx dy \\ &= \frac{1}{8} R^4 - \left(\frac{\pi}{4} R^2\right) \left(\frac{4R}{3\pi}\right) \left(\frac{4R}{3\pi}\right) \\ &= \left[\frac{1}{8} - \frac{4}{9\pi}\right] R^4 \end{aligned}$$

$$\bar{I}_{x'y'} = -0.01647 R^4$$

Agrees w/ Flori crib sheet:

$$\bar{I}_{x'y'} = (-1)^n \cdot 0.01647 R^4$$



This orientation is Neg.